must use VCE ADTs and terminology

VIRTUAL SCHOOL VICTORIA

2026

**NAME:**

# ALGORITHMICS UNIT 3

## SAC 1: Abstract Data Types (Weeks 1 to 4)

## Outcome 1

## Date of Completion: 2-6 March 2026

**Reading Time: 5 minutes**
**Writing time: 55 minutes**
**TOTAL (60 minutes)**
QUESTION AND ANSWER BOOK

| *Type* | *Number of questions* | *Number of questions to be answered* | *Number of marks* |
|---|---|---|---|
| Short/Extended Response | 9 | 9 | 40 |
| | | Total | 40 |

**Materials supplied**
- Question and answer book of 10 pages

**Materials permitted**
- Pens/Stationary

**No calculator or reference material permitted.**

**Instructions**
- Write your **name** in the space provided above on this page.
- All written responses must be in English, point form is preferred.

**Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices.**

1.

a. Define a tree.

A connected acyclic graph (both needed)

(1 mark)

b. What possible numbers of edges could a tree with 10 vertices have?

9

(1 mark)

Q. Give one similarity and one difference between a set and a dictionary ADT.

Similarity - unordered. (could say "lookup using hash")

Difference - dictionaries store values assigned to keys, sets really just store the keys

lots saying dicts are ordered

(2 marks)

2.

(a) The following items are added to a minimum priority queue. The priority is given in brackets.

Added 1st: Alpha (2)

Added 2nd: Bravo (1)

Added 3rd: Charlie (1)

Added 4th: Delta (2)

Write down the order in which they are dequeued, from first to last.

B C A D

(2 marks)

(b) Write down the signature for a method to add an element to a priority queue.

enqueue: queue x element x int --> queue

(2 marks)

3. Consider the following pseudocode:

```
a = 2
b = 10
if a > b:
  c = 0
  d = b
  while a > d:
    d = d * b
    c = c + 1
  return c
else:
  swap values of a and b
  c = 0
  d = b
  while a > d:
    d = d * b
    c = c + 1
  return c
```

a b c d
2 10
10 2 0 2
1 4
2 8
3 16

a. Show what the code returns using the values given.

3

(2 marks)

b. By annotating the code, show how modularisation can be used to improve the code, stating clearly one advantage of your approach.

Indicated part can be modularised, to avoid repetition/errors/improve readability (need a reason)

(2 marks)

4. Find the values of A, B and C which make the following Boolean expression False.

```
{not (A and B)} or (A and C)
```

| A | B | C |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |
| 1 | 1 | 1 |

can deduce

True, (A nand B)

True A nand C

(2 marks)

5. Bruce is constructing a model to organise his music collection. The table below shows the kind of information that needs to be stored:

```
Song name, Artist Name, Genre, Play Count, Length
```

He wants to be able to quickly look up songs either by artist or by genre. Initially he does this using a list of dictionaries, where each song is represented by a dictionary.

```
song_list = [

…

{name: "Taking The Queen", artist="Bruce Dickinson",
genre="Heavy Metal", play count=15, length="4.48"}

…

]
```

(a) Explain two distinct reasons that this combination of ADTs is suboptimal.

Has to cycle through the list to find all songs of a particular artist or genre.

Use of dictionary to repeat titles increases storage requirements

(3 marks)

(b) Construct your own combinations of ADTs to solve his problem. Clearly explain why your solution allows efficient lookup of songs by both artist and genre.

One example:

A list of songs, where each song is stored as an array containing e.g. title, album, artist, genre. Also add an ID.

A dictionary of genres, where the key is the genre and the value is a list of IDs.

A dictionary of artists on the same basis

Dictionaries allow fast lookup of all the songs for any particular artist/genre.

(5 marks)

6. Consider the following pseudocode

```
a = 30
b = 12
c = 1
d = 1
while d > 0:
  c = a // b
  d = a % b
  a = b
  b = d
return b
```

a. What does the pseudocode return when run with the given values?

0 (Infinite)

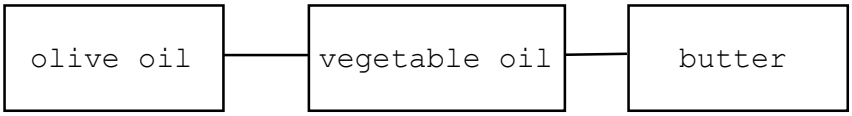
(2 marks)

b. Circle the line(s) of pseudocode which could be removed without changing the overall working of the algorithm.

(2 marks)

7. Jayden wants to create an AI system which can help you to complete recipes with missing ingredients by indicating which ingredients can be substituted for others.

He starts with this graph, indicating that butter or olive oil can both be used instead of vegetable oil.




a. This erroneously leads to somebody trying to pour butter over a salad instead of olive oil. Explain why this happened, and how the graph can be changed to avoid this problem.

The graph is transitive (or words to that effect)

Have directed edges to show that e.g. oil can be replaced with butter. (other options available)

(2 marks)

b. Under what situation might you decide to use weights on the edges of the graphs?

Many options - e.g. compatibility (must be of the relationship between two ingredients, not e.g. cost of butter)

(2 marks)

c. Explain how we could further modify the model to indicate that, for example, olive oil can be substituted for butter in savoury dishes but not sweet ones.

not make a new model

Multiple graphs

(decision tree = 0m)

Edges with properties

(2 marks)

8. You are to use ADTs to construct a model which could be used to find a solution to the 'jealous husbands' problem with four couples: There are four married couples on one side of a river. They have a boat which carries two people. At least one person must be in the boat to cross the river. The husbands are all jealous, such that they will not allow their wives to be in the presence of another man, unless they themselves are present. That means that if a man's wife is on one side of the river, then either he must be on the same side, or there must be no other men on that side.

a. Explain how each state of the problem can be represented using ADTs.

need boat

A state graph

need to show individual ppl (not just "4 women")

~~Each node stores a list of people on one side, a list on the other, and a value to indicate the position of the boat (others possible)~~

(2 marks)

b. Show how your combination can be used to represent the initial state.

[A B C D 1 2 3 4]
[] left

(1 mark)

c. Show two further nodes which can be achieved from the initial state within the constraints of the problem.

[A B C D 1 2]
[3 4]
Right

[A B C D 3 4]
[1 2]
Right

etc.

(2 marks)

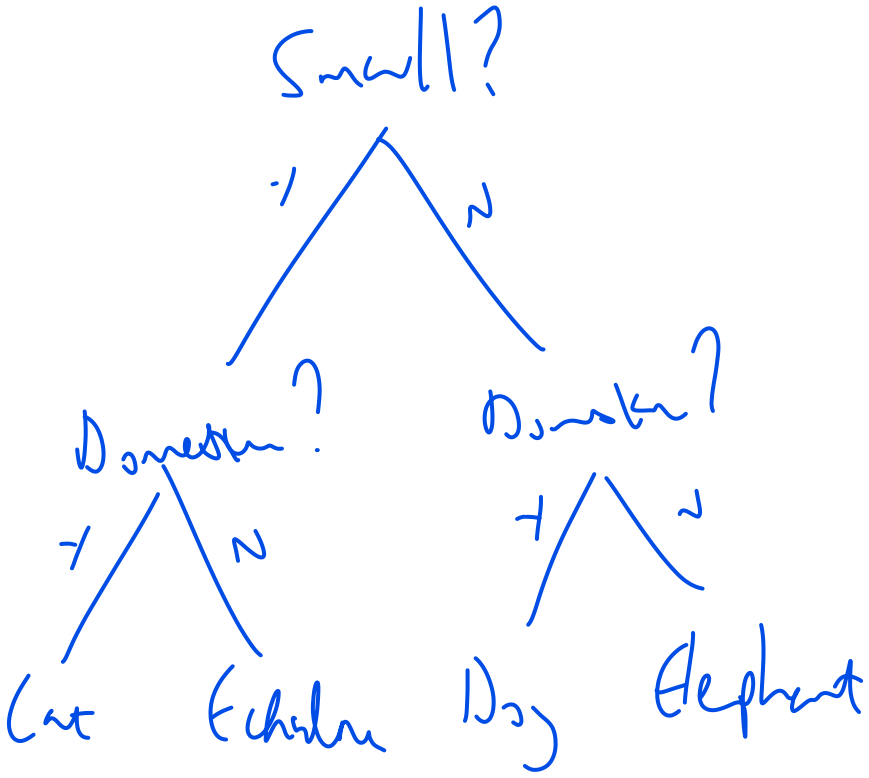
9. Construct a decision tree to distinguish the following animals using only Yes-No questions, using the properties of the animals. Your tree must have a maximum of three nodes, and asks only two different questions. (i.e. two nodes will ask the same question). You must use correct notation.

Cat – Small – Domestic

Dog – Medium – Domestic

Echidna – Small – Wild

Elephant – Large – Wild




(3 marks)

**STUDENT DECLARATION OF AUTHENTICITY**

This SAC is my own work and has been completed under the stated requirements.

Student Signature: Date:

Supervisor comments: